Integrated Quantum Dot Lasers for Parallelized Photonic Edge Computing

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Abstract. Scalable photonic computing for edge AI remains limited by the inefficiency of integrated light sources. Here, we present a photonic computing unit powered by a quantum dot mode-locked laser (QD-MLL) heterogeneously integrated on Intel's 300 mm silicon photonics platform. Our approach bridges device-level demonstrations with link- and system-level evaluations, experimentally validating core computing functions. Compared with hybrid light sources, the isolator-free QD-MLL reduces front-end optical loss by 2–6 dB. When combined with a co-designed computing architecture, the optical link achieves over 10 dB improvement in signal-to-noise ratio and more than 50% reduction in thermally induced weight drift. Under realistic laser structure and power constraints, the system delivers 1.7× higher effective scalability, yielding >40% enhancement in computational density and ~30% gain in energy efficiency. To further enable robust deployment at the edge, we introduce a hardware-aware training framework that strengthens inference reliability while minimizing hardware overhead. Collectively, this work establishes a cross-layer framework—from device to architecture to algorithm—toward a unified hardware-software strategy for scalable photonic edge computing.

Keywords: Quantum dot laser, photonic computing, heterogeneous integration, edge device

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1 Introduction

Edge computing brings computation closer to data sources, enabling low-latency AI processing essential for Internet of Things (IoT) applications[1,2]. However, stringent hardware constraints—particularly in power and form factor—make it challenging to support complex AI models on edge devices. A key computational bottleneck lies in matrix-vector multiplication (MVM), the core operation in neural inference that imposes high energy costs and latency, driving the need for specialized accelerators.

Photonic computing offers a promising alternative with its inherent advantages in bandwidth, latency, and parallelism, particularly when implemented on mature silicon photonics platforms[3,4]. However, silicon's low emission efficiency remains a major barrier to integrated photonic systems[5,6]. To address this, companies such as Ayar Labs employ hybrid integration, coupling separately packaged III-V lasers to silicon photonic integrated circuits (PICs) via optical interfaces[7]. While modular and relatively straightforward, hybrid approaches introduce challenges in packaging complexity and thermal management.

In contrast, Intel's heterogeneous integration platform supports large-scale manufacturing while improving coupling efficiency and thermal uniformity. In 2024, Intel demonstrated a prototype optical compute interconnect (OCI) chiplet with integrated light sources, confirming the feasibility of scalable, fully integrated photonic computing hardware[8]. Nonetheless, most heterogeneous demonstrations still rely on quantum wells (QWs) as the gain medium. QWs are highly sensitive to back-reflections and typically require bulky optical isolators[9,10]. In addition, their intrinsic gain characteristics result in severe mode partition noise, limiting their suitability as multi-channel sources for wavelength-division multiplexing (WDM) systems[11,12].

Quantum dots (QDs) offer a compelling alternative. Their near-zero linewidth enhancement factor provides high tolerance to back-reflections, allowing for an isolator-free configuration. Moreover, QDs inherently suppress mode partition noise, supporting stable multi-channel emission in mode-locked operation. Collectively, these advantages make heterogeneously integrated QD mode-locked lasers (QD-MLLs) a scalable and energy-efficient WDM light source[13,14]. However, the system-level benefits of QD-MLLs in photonic computing remain largely unexplored, and their successful transition to heterogeneous platforms requires cross-layer optimizations[15].

In this work, we present the design of a photonic computing unit (PCU) driven by a heterogeneously integrated QD-MLL on Intel's 300 mm silicon photonics platform. The study combines device-level demonstrations with link- and system-level evaluations, supported by experimental validation of fundamental computing functions. Compared with hybrid light source schemes, the heterogeneous QD-MLL reduces front-end optical loss by 2–6 dB, owing to seamless coupling and isolator-free integration. When paired with a co-designed computing architecture, the optical link exhibits more than 10 dB improvement in signal-to-noise ratio (SNR) and over 50% reduction in thermally induced weight drift. System-level analysis under fabrication and power constraints shows a 1.7× scalability boost, translating into >40% higher computational density and ~30% increase in energy efficiency compared to an otherwise identical system using hybrid multi-channel light sources. Such enhancements are broadly applicable across other WDM-based photonic computing architectures. To support robust deployment in edge scenarios, we additionally introduce a general hardware-aware training framework (HATF) that improves inference reliability while minimizing hardware overhead. This study establishes a cross-layer methodology—spanning device, architecture, and algorithm levels—for on-chip QD-laser-driven photonic edge computing. It provides a unified hardware–software co-design strategy for advancing scalable and energy-efficient edge AI on a heterogeneous silicon photonics platform.

2 QD-MLL-driven PCU on a heterogeneous integration platform

Fig. 1a illustrates the concept of an on-chip QD-MLL-driven photonic computing system, where the PCU is co-packaged with electronic integrated circuits (EICs) to enable chiplet-scale OCI implementation. This system relies on the seamless integration of multi-channel laser sources, modulators, weight arrays, and photodetectors (PDs), supporting wafer-scale batch fabrication of fully functional photonic chips in a single run[11,16]. By eliminating fragile chip-to-chip optical input/output (I/O) interfaces, the architecture strengthens system robustness and reliability, simplifies packaging, and supports higher-speed interconnects[17]. In addition to the photonic front-end, the system integrates electronic components for signal transmission and control. The transmitter includes a signal generator (SG) and a digital-to-analog converter (DAC), while the receiver consists of a trans-impedance amplifier (TIA) and an analog-to-digital converter (ADC). Local random-access memory (RAM) stores intermediate data, and an embedded microcontroller unit (MCU) manages configuration and system control.

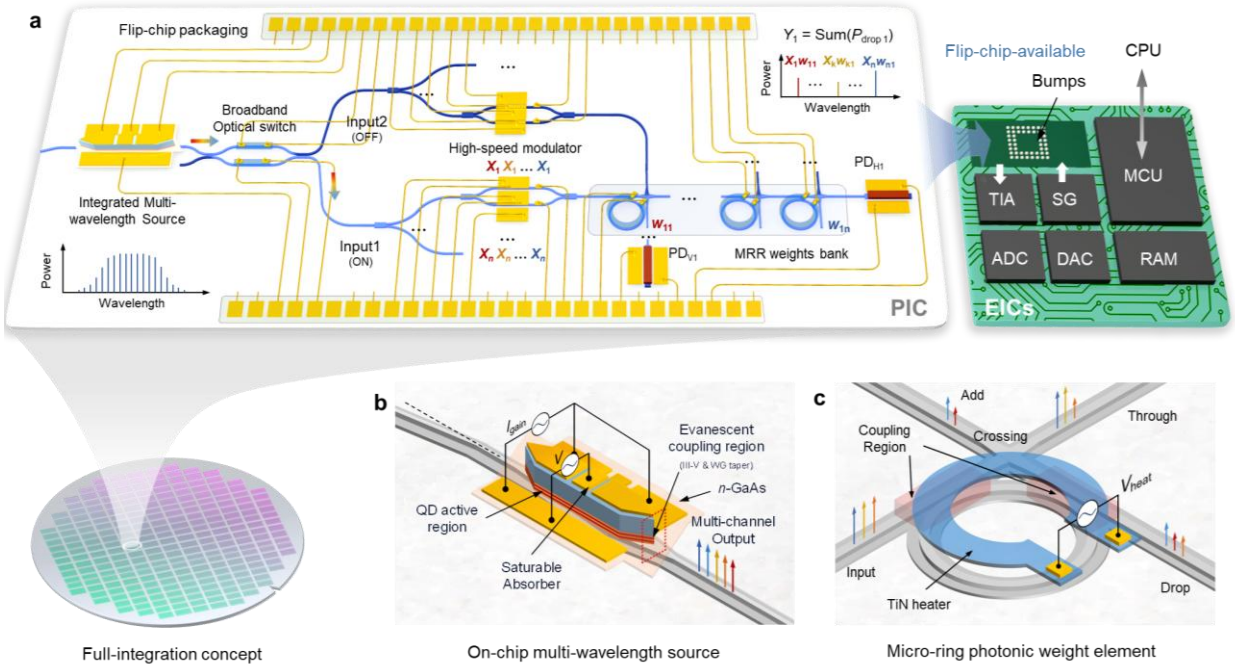

Fig. 1 Integrated QD-MLL-driven photonic computing unit (PCU). a Conceptual illustration of the fully integrated PCU-chiplet system. A heterogeneous QD-MLL provides multi-wavelength carriers, modulated by TWMZMs, and weighted by a crossbar-configured MRR array. Output signals are detected by PDs, and peripheral electronics are co-packaged via flip-chip integration. **b** Heterogeneously integrated QD-MLL with saturable absorber and efficient III-V to silicon coupling. **c** MRR-based weight mapping element with crossbar configuration.

A stable on-chip light source is enabled by a heterogeneous platform that supports flexible gain material selection and high-throughput manufacturing. The vertical stack achieves precisely-aligned evanescent coupling between the III-V gain region and silicon waveguides, minimizing coupling loss (<0.5 dB[16,18]) and removing the active facet alignment required in butt-coupled designs[5]. The incorporation of QDs as the gain medium further enhances the laser's tolerance to optical back-reflection and enables stable multi-wavelength mode-locking. This behavior stems from the exceptionally low linewidth enhancement factor of QDs ($\alpha < 0.3$)[19], which suppresses refractive index fluctuations and mitigates coherence collapse, allowing stable operation under optical feedback levels up to -7 dB[9]. In addition, the superior gain characteristics and ultrafast carrier dynamics of QDs facilitate short-cavity mode-locking, enabling more compact laser designs and higher repetition rates[20]. The schematic of our heterogeneously integrated QD-MLL is shown in **Fig. 1b**. Compared with conventional schemes based on nonlinear optical processes[21,22] or discrete laser modules[23,24], this approach simplifies control while improving both

energy and area efficiency. A detailed comparison of representative light source strategies is provided in **Tab. S1**. Fully exploiting the integration and scalability potential of on-chip lasers requires a computing architecture co-designed to flexibly adapt to different optical channel counts with minimal hardware overhead. Specifically, this requirement shapes the design of the PCU, influencing channel-to-core mapping strategies, wavelength routing, and interconnect topology. Accordingly, we employ microring resonators (MRRs) as the fundamental units for weight implementation, owing to their compact footprint, low power consumption, and dense WDM compatibility[25]. The MRRs adopt a crossbar-configured topology (**Fig. 1c**), which avoids the closed-loop feedback paths of cascaded or serpentine arrays, thereby reducing parasitic reflections and improving laser stability[26,27]. This layout supports modular scaling into two-dimensional arrays, while its bidirectional connectivity mitigates path-length mismatches and timing skew across channels. Moreover, crossbar configurations with symmetric layout can support in-situ transposition of MRR weight matrices, facilitating hardware reuse across varying computational stages without physical reconfigurations[28].

3 Photonic device co-design and functional validation of the PCU

Efficient optical transition in heterogeneous integration demands strong mode overlap between the III-V gain layer and the silicon waveguide. To achieve this, we adopt an optimized silicon-on-insulator (SOI) configuration and designed dedicated coupling structures (see **Supplementary Note 2**)[29]. The fabrication process, shown in **Fig. 2a**, begins with 12-inch SOI wafers produced on Intel's 300 mm complementary metal-oxide-semiconductor (CMOS) line, which are subsequently cored down to 4-inch for wafer bonding and standard laser processing in a university cleanroom[16,30]. Parametrically optimized linear tapers in both the III-V and silicon layers ensure spatial mode matching across the bonding interface. The taper geometries and corresponding mode evolution are shown in **Fig. 2a(i–iii)**.

Instead of relying on cleaved facets, a pair of silicon-based loop mirrors is implemented to define the laser cavity and provide optical feedback. Within this cavity, saturable absorbers (SAs) are selectively introduced in the III-V/silicon gain section to provide electrical isolation and to segment the cavity for harmonic mode-locking. The total round-trip optical length, including the III-V gain section, silicon feedback paths, and SA-defined harmonic order, is jointly optimized to meet the target repetition rate while maintaining adequate gain and channel spacing. **Fig. 2b** shows

a representative QD-MLL array fabricated on the heterogeneous integration platform. To quantify the performance benefits of QD-MLL integration within the PCU, we developed a numerical model to evaluate the impact of light source characteristics on the SNR per computing link. With both the QD gain medium and the heterogeneous integration platform, the seamless light source reduces coupling loss and eliminates the need for external fiber-based isolators or amplifiers, cutting front-end power loss by 2–6 dB compared to hybrid QWL-based multi-wavelength sources. The low relative intensity noise (RIN) of QDs[19], due to their discrete energy levels and suppressed carrier fluctuations, together with higher per-channel optical power, leads to an overall SNR increase of exceeding 10 dB (**Fig. 2c**). This advantage supports higher signal integrity and computational accuracy.

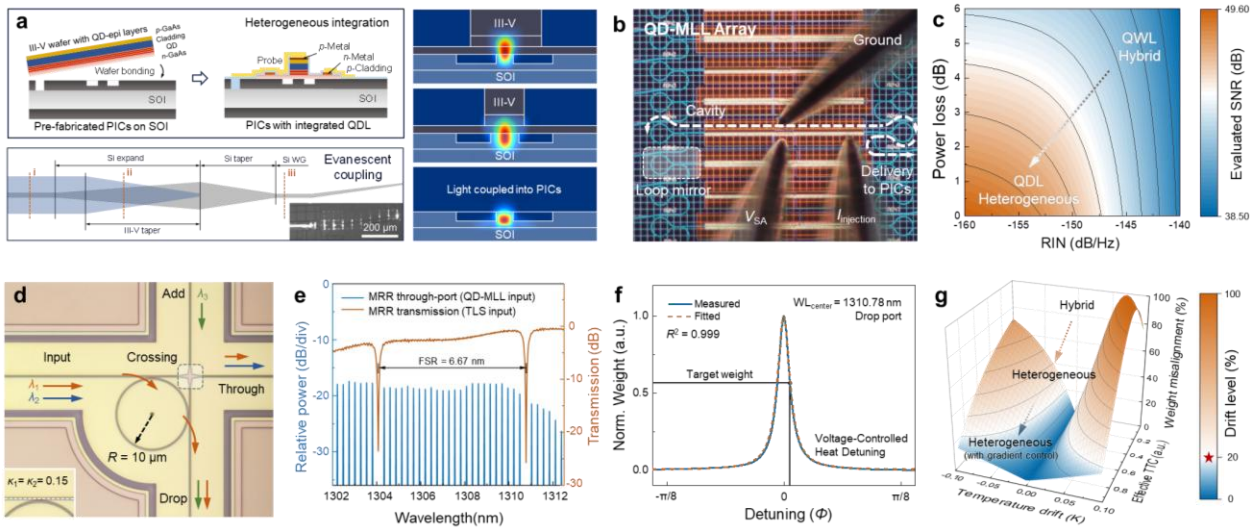

Fig. 2 Design and characterization of heterogeneously integrated photonic components. a Schematic of the QD-MLL fabrication process, including wafer bonding, taper design, and evanescent coupling from III-V to silicon (inset: infrared image of the coupling region; right: simulated mode transitions). **b** Optical micrograph of the fabricated QD-MLL device array with annotated injection, cavity, and output regions. **c** Evaluated SNR as a function of RIN and power loss, comparing heterogeneous QD laser and hybrid QW laser sources. **d** Optical image of a crossbar-configured MRR with labeled ports and coupling ratio. **e** Measured MRR through-port spectra under QD-MLL and TLS inputs, showing >15 dB extinction. **f** Drop-port response and fitted model (R^2 = 0.999) for weight mapping. **g** Simulated weight drift (at maximum slope point, w=0.5) under typical TEC fluctuation, showing over 50% suppression of weight thermal drift with heterogeneous integration.

The fabricated crossbar-configured MRR on Intel's heterogeneous platform is shown in **Fig. 2d**. As a proof-of-concept demonstration of functional feasibility, optical channels from the on-chip QD-MLL were directly coupled into a discrete MRR-based weight element without the use

of external components. Experimental results are shown in **Fig. 2e** with detailed measurements and function evaluations provided in **Supplementary Note 3**. Using a tunable laser source (TLS, with isolator) as input, the baseline transmission spectrum reveals clear resonance dips with a free spectral range (FSR) of 6.67 nm. When switching to the QD-MLL (without isolator and amplifier), a WDM comb spectrum is observed. By finely adjusting the heater voltage, individual MRR resonances align with specific laser lines, yielding extinction ratios exceeding 15 dB. The QD-MLL employed in this demonstration provides ~50 GHz channel spacing, accommodating over 20 optical channels across the MRR FSR with channel-to-channel variation below 3 dB—well-suited for dense WDM-based computing applications. The drop-port transmission, shown in **Fig. 2f**, represents the effective weight mapping profile after normalization, with the aligned state in **Fig. 2e** corresponding to the maximum mapping condition ($w = 1.0$). Adjusting the resonance detuning via heater voltage allows continuous control over the mapped weight. Finally, the measured resonance spectrum is accurately fitted using a coupled-mode theory model ($R^2 = 0.999$), providing the foundation for system-level modeling and look-up table (LUT) construction.

The spacing between adjacent MRRs is optimized to balance layout compactness and thermal crosstalk. While thermal isolation and thermoelectric cooler (TEC) regulation can suppress immediate thermal fluctuations, residual long-term temperature variations may still induce gradual weight drift. To evaluate the robustness under such disturbances, we introduce a temperature tracking coefficient (TTC), which quantifies the spectral drift between the laser and the MRR based on their thermal tuning characteristics. An analytical model is then established to translate the spectral mismatch into quantitative weight deviation. The evaluation is conducted at the most thermally sensitive bias point (weight = 0.5), using the fitted MRR model (**Fig. 2f**) and typical TEC fluctuations of ±0.05 °C. As illustrated in **Fig. 2g**, heterogeneous integration improves TTC by using a shared substrate that unifies thermal environment and control. This avoids mismatches between multiple TECs and reduces weight error by over 50% compared to hybrid integration.

Furthermore, the enhanced thermal predictability of this platform allows pre-configured temperature gradients within the TEC, enabling tighter control. Under ideal conditions, TTC can reach its theoretical limit, maintaining weight alignment within 10% error. All thermal analyses mentioned above are detailed in **Supplementary Note 4**. The travelling-wave Mach-Zehnder modulator (TWMZM) and PD within the computing link were similarly designed to match the

PCU's input and output requirements, achieving energy-efficient and high-speed operation (theoretically >28 GHz). **Supplementary Note 5** provides detailed design and simulation data.

4 Model-based scalability and precision evaluation under laser constraints

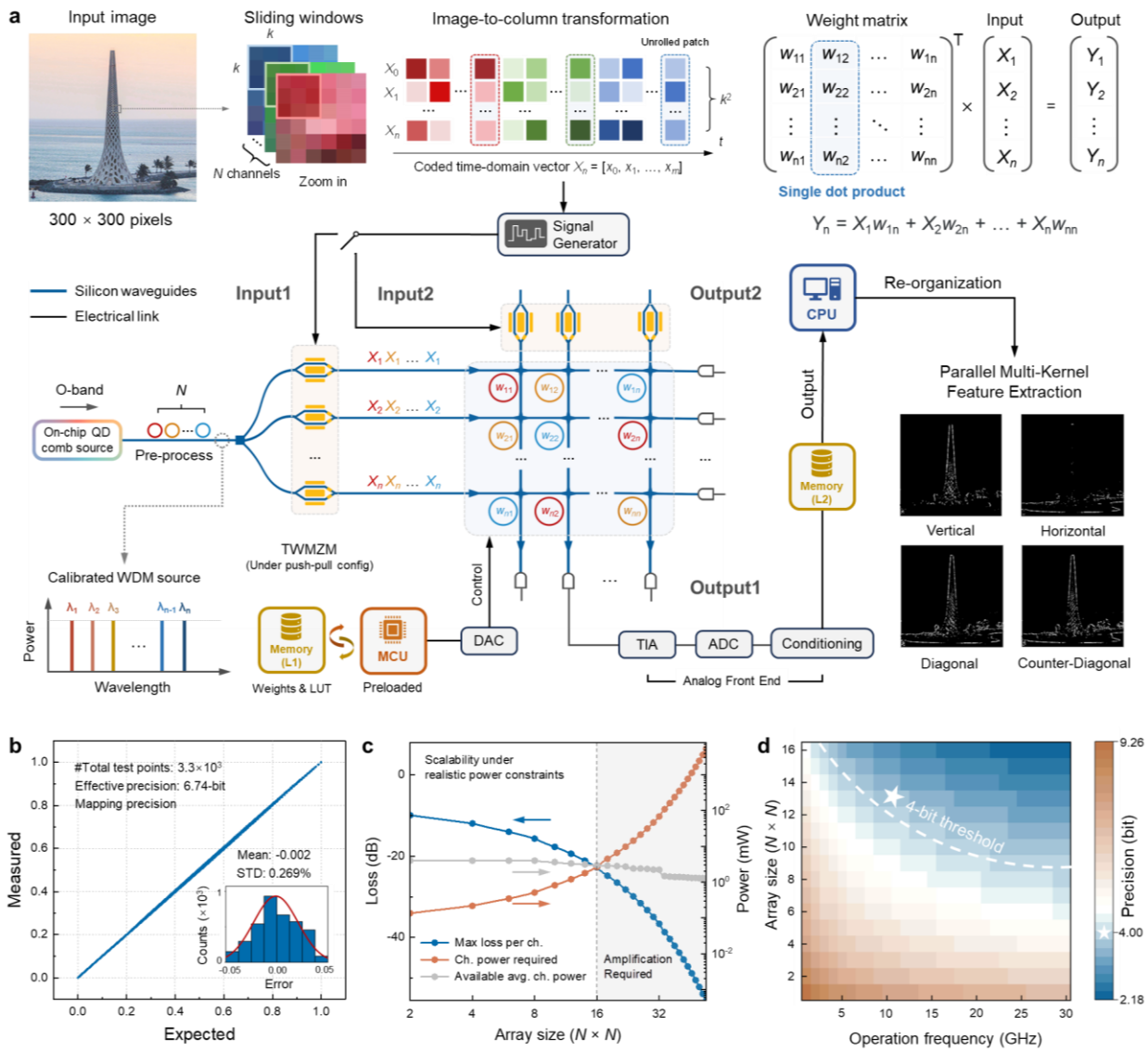

Fig. 3 Computational scalability and precision analysis under laser structure and power constraints. a Schematic of the proposed optoelectronic computing flow, illustrating the image encoding scheme, parallel MVM execution via the MRR crossbar, and feature extraction for multi-kernel convolution. **b** Weight mapping precision of a single MRR, achieving 6.74-bit equivalent operational resolution. **c** Scalability and **d** achievable end-to-end precision analyzed using a system-level optoelectronic model under hardware constraints.

Building on the optimized device and architecture designs, along with prior experimental results, we developed a hierarchical model-based analytical framework to evaluate both the scalability and precision of the PCU-chiplet system. This framework considers laser-induced limitations (i.e., QD-MLL design envelope and per-channel power budgets) and hardware penalties (i.e., link-level losses and optoelectronic noise), enabling realistic and quantitative evaluations. The workflow of the QD-MLL-driven PCU-chiplet system is illustrated in **Fig. 3a**. Input data are first encoded into kernel-compatible (k^2) vectors and converted into time-domain voltage sequences. These sequences are then routed through lateral or vertical ports and simultaneously modulated onto N optical channels generated by the QD-MLL via TWMZMs. The modulated signals are injected into the MRR crossbar array programmed with target weights. This array simultaneously performs multiplications and intrinsic wavelength-to-channel mapping, eliminating the need for scaling-limited demultiplexers. As shown in **Fig. 3b**, a representative MRR demonstrates an effective mapping precision of 6.74 bits, which reflects the static configuration resolution prior to runtime operations. Finally, the PDs sum the outputs from the N wavelength channels within each row to complete the MVM. The computed results go through analog front-end processing and are finally output to the digital domain for reorganization.

A numerical model was constructed to estimate the per-channel power output of QD-MLLs by exploring parameter combinations of base repetition rate and harmonic order within fabrication-constrained ranges. In parallel, we estimate the required per-channel power by accounting for cumulative photonic link losses, including coupling, modulation, and routing penalties (see **Tab. S2**). The effective scalability of the system is defined by evaluating, for each array size, whether the available output power exceeds the minimum required threshold, which is determined by cumulative link losses and PD's sensitivity. The maximum feasible number of directly supported MRR channels is thus taken as the scalability limit of a single laser. Benefiting from reduced power loss, the QD-MLL provides higher per-channel optical power and supports up to N = 16 MRR channels without amplification (**Fig. 3e**), representing a 1.7× effective scalability improvement over a hybrid light source.

Within the evaluated hardware scale, we further constructed an optoelectronic link model that captures noise from modulation, transmission, detection, and electrical readout. Based on this model, we derive the upper bound of achievable precision for different (array size, frequency) configurations by accounting for signal attenuation, accumulation-induced bit-width expansion,

and frequency-dependent noise amplification. As shown in **Fig. 3d**, a two-dimensional heatmap illustrates the calculated bit precision. The dashed contour corresponds to the 4-bit threshold, commonly accepted as the minimum requirement for standard neural network inference, thus defining the feasible design region. Detailed modeling procedures and assumptions in this framework are provided in **Supplementary Note 6**.

5 Generalized hardware-aware optimizations of the algorithm

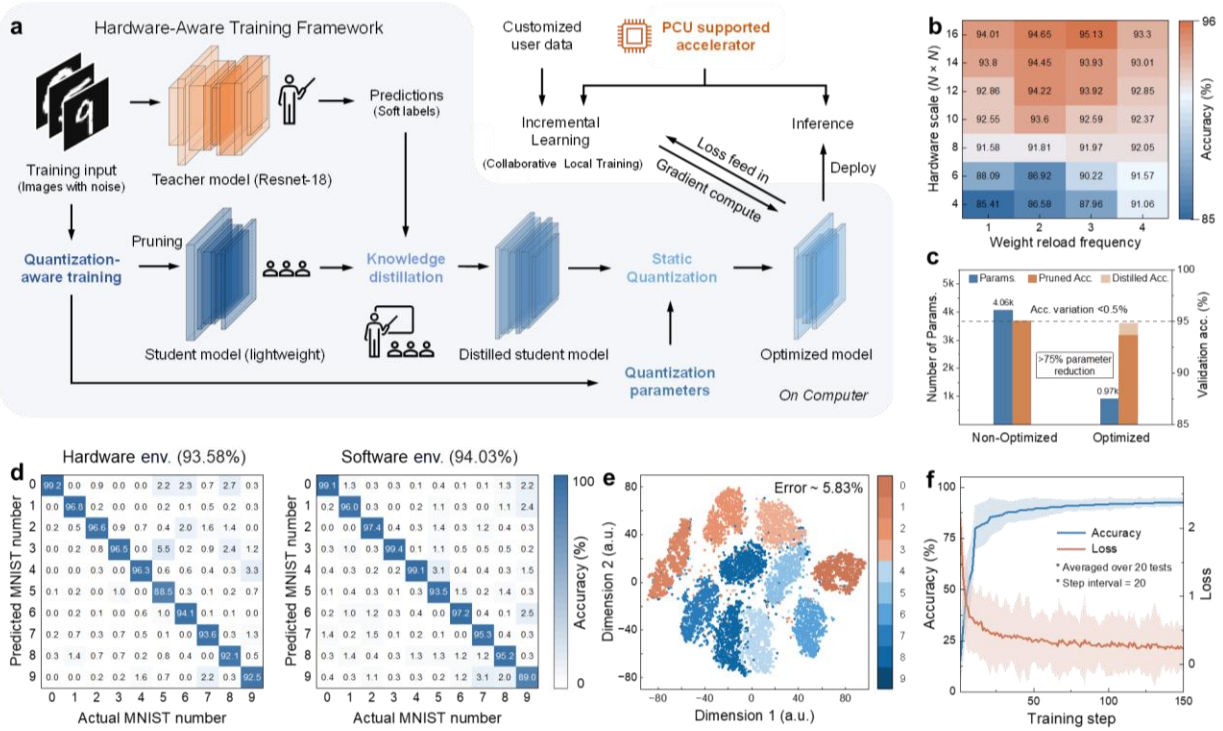

Fig. 4 Validation of hardware-aware training framework (HATF). a Schematic of the HATF, combining quantization-aware training, pruning, and knowledge distillation for deployment on PCU with support for user-specific incremental learning. **b** Heatmap of model accuracy under varying hardware configurations (array scale and weight update counts). **c** Parameter reduction and validation accuracy comparison before and after HATF optimization, showing >75% compression with <0.5% accuracy drop. **d** Confusion matrices from the generalization test under different inference environments, showing less than 1% performance gap. **e** *t*-SNE projection of the collaborative model's classification performance. **f** Training trajectory showing accuracy and loss convergence.

To adapt neural network models to the hardware limitations of PCUs, we introduce a general HATF that incorporates hardware constraints into each stage of the model training process (**Fig. 4a**). The pipeline begins with low-bit quantization and link-calibrated noise injection, simulating deployment-related non-idealities during training. We then explore the design space under resource

constraints, varying architectural parameters to identify configurations that balance cost and accuracy. This yields sparse networks with fewer parameters, reducing storage and updating overhead while maintaining accuracy.

The optimized architecture is subsequently passed to a knowledge distillation stage, where a pretrained teacher network provides soft targets to supervise the sparse student model, compensating for the potential accuracy loss[31]. PCU's intrinsic support for in-situ matrix transposition enables the same weight banks to be reused for both forward and backward propagation. This ability allows the system to support collaborative incremental learning, wherein fine-tuning can be locally performed with external assistance. Through this mechanism, the deployed model can be updated in response to evolving environments or application-specific requirements, thereby prolonging system lifetime and enhancing flexibility.

We demonstrate the effectiveness of HATF through a case study using a multi-layer perceptron (MLP) trained on the MNIST database[32]. As shown in **Fig. 4b**, classification accuracy is primarily influenced by two hardware configurations: the crossbar array size and the frequency of weight reprogramming, which together define the system scalability and update overhead. Building on this, we performed a fine-grained search across 421 candidate architectures (see **Supplementary Note 7**), jointly adjusting model structures and pruning levels. Further incorporating knowledge distillation with a ResNet-18 model as the teacher, the student model can be compressed to 974 parameters—an 80% reduction compared to the baseline MLP—while maintaining validation accuracy (**Fig. 4c**). The generalization test on a held-out dataset under simulated hardware conditions shows a degradation of less than 1% in model recognition accuracy compared with using ideal software environments (**Fig. 4d**).

To evaluate the local adjustability of the PCU, we simulate training-phase operations by executing both forward ($W \cdot x$) and backward ($W^{\mathrm{T}} \cdot \delta$) MVMs under previously defined hardware constraints. Loss computation and weight updates are idealized to reflect a hybrid host–device collaborative scenario. As illustrated in the *t*-distributed stochastic neighbor embedding (*t*-SNE) plot (**Fig. 4e**)[33], the trained model preserves class separation with an average distributional error of ~5.83% compared to the ground truth. Training curves, an average of over 20 runs, confirm the stability and convergence of the adaptation process (**Fig. 4f**). The pseudocode for implementing this framework is provided in the **Supplementary Appendix**.

6 Performance evaluation for a co-packaged chiplet system

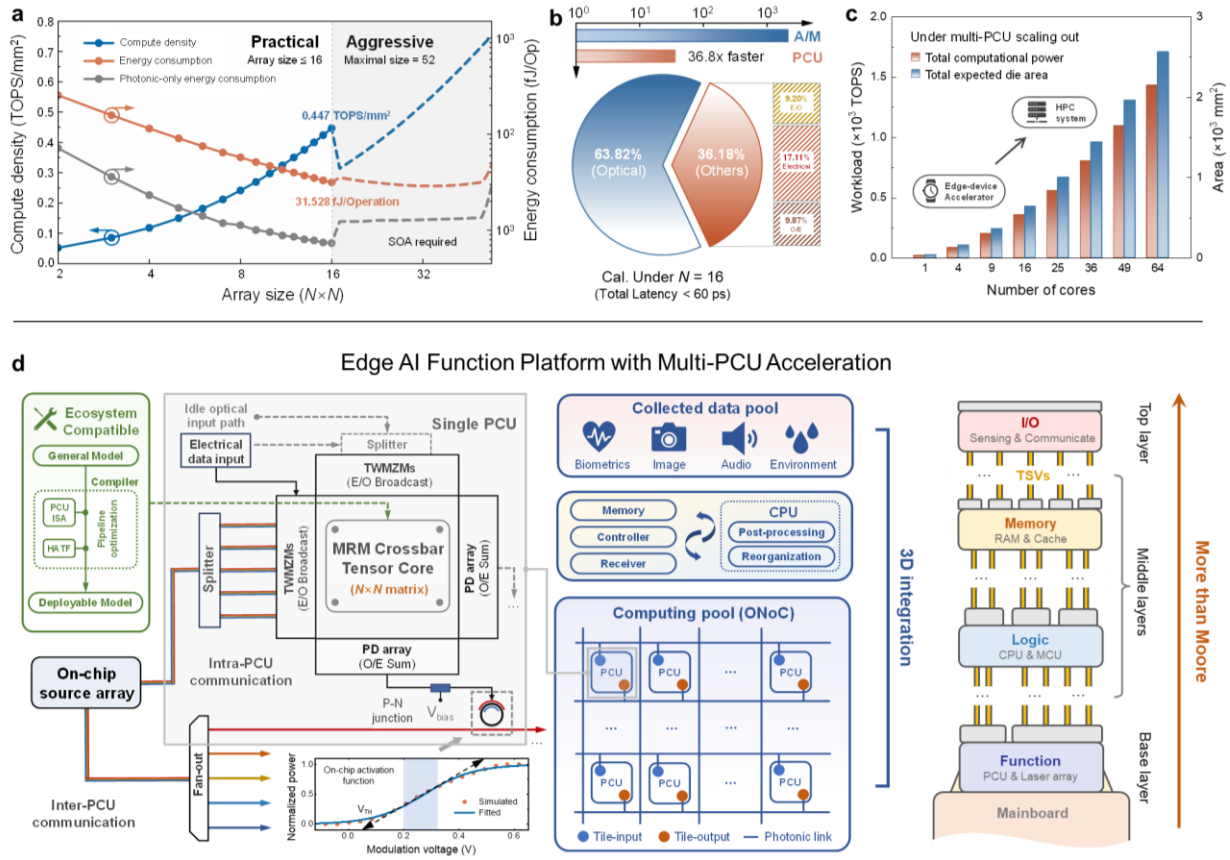

Fig. 5 PCU system-level scaling performance and multi-core parallelization strategy. a Compute density and energy consumption as functions of array size under practical and aggressive scaling scenarios. **b** Compute latency and breakdown analysis of a 16 × 16 MVM task, showing 36.8× speedup compared to electronic baselines. **c** Projected performance and area of multi-PCU systems under parallel scaling-out configurations. **d** Multi-PCU enabled edge AI platform supporting efficient parallel execution via on-chip optical interconnects and heterogeneous 3D integration.

To quantitatively evaluate the computational performance gains of heterogeneous QD-MLL integration in PCU-chiplet systems, we developed a system-level analytical performance evaluation model. Guided by prior physical and link-layer analyses, we determined compatible PIC and EIC components and extracted typical operating parameters. The corresponding specifications, including die area and energy consumption breakdowns, are summarized in **Tab. S3** and **Tab. S4**.

Building upon these inputs, we calculated the computational density and energy consumption of the system across distinct array sizes, as illustrated in **Fig. 5a**. For a practical array size ($N \leq 16$), the system delivers a peak computational density of 0.447 TOPS/mm^2 and an energy

consumption of 31.528 fJ/operation. With identical PCU architecture, replacing the hybrid multi-channel source with the QD-MLL boosts computational density by up to 40% and energy efficiency by 30.33%. This gain is consistently validated across representative WDM-based photonic computing systems, with detailed calculations and comparisons provided in **Tab. S5**. In addition, we analyzed system latency as a complementary metric. At the maximum practical scale ($N = 16$), the high parallelism of the WDM operations compresses the effective MVM latency to less than 60 ps. Compared to a 7 nm CMOS-based advanced electronic accelerator using a standard adder/multiplier (A/M) structure, this results in a >36.8× latency reduction for equivalent parallel MVM computation (**Fig. 5b**).

To evaluate scaling beyond the single-laser limit, we considered more aggressive configurations with higher channel counts ($N > 16$). Based on the measured FSR and Q-factor of the MRRs, the theoretical maximum number of supported WDM channels is estimated to be 55. Within this defined range, system performance was evaluated using previously demonstrated QD-based semiconductor optical amplifiers (SOAs)[34]. The results are shown in the dashed region of **Fig. 5a**. The initial inclusion of SOAs causes a slight performance drop, but as the array size increases, their additional power and area overhead are offset by scaling benefits, enabling continued advances in computational density and energy efficiency. For $N > 50$, SOA saturation reduces efficiency and defines the system's scaling limit.

Furthermore, the compact footprint and low power consumption of QD-MLLs facilitate scalable on-chip source arrays. Combined with the PCU's intrinsic compatibility with optical interconnects, this enables multi-core scale-out configurations where each PCU functions as a modular computing tile, similar to modern parallel processors. Each PCU tile maintains high compute density and energy efficiency, while cooperation across tiles achieves exponential gains in aggregate workload, as shown in **Fig. 5c**. As the number of PCU cores increases, the system supports distributed execution of large-scale models, with subtasks allocated to individual PCUs and processed in parallel. By offering configurability in both intra-PCU array size and inter-PCU core count, the architecture scales flexibly across diverse application scenarios, ranging from edge inference to high-performance computing (HPC). The methodology and modeling details for the analyses in this section are described in **Supplementary Note 8**.

7 Discussions and Conclusions

The complete design of a parallel multi-core computing system, supported by the proposed PCU architecture, is conceptually illustrated in **Fig. 5d**. Modular computing tiles can be assembled without redesigning optical interfaces or waveguide routing, and, through the optical network-on-chip (ONoC)[35], they are seamlessly interconnected to form a reconfigurable computing fabric. The integrated laser source array concurrently supplies WDM channels for both intra-tile and inter-tile communication, thereby improving spectral utilization and overall system scalability. Within each PCU, intermediate results are periodically transferred to photonic links (i.e., waveguides) via high-speed MRR modulators. In addition to this communication role, the same modulators serve as on-chip activation elements, owing to the inherent nonlinear characteristics of PN-junction-based electro-optic modulation[22].

To support programmability, a domain-specific instruction set architecture (ISA) can be introduced to define hardware-level operations such as modulation, routing, and matrix transposition[3]. Encoded as addressable instructions, the ISA provides a structured interface for computation and communication across tiles, enabling fine-grained task scheduling, runtime adaptation, and dynamic control of photonic elements[36,37]. Building on this abstraction and combined with HATF, a dedicated compiler toolchain can map high-level AI models onto hardware-executable flows, incorporating model quantization, photonic path planning, and pipeline-aware task decomposition for efficient deployment of photonic hardware.

Beyond the photonic computing pool, building a fully functional platform requires peripheral electronics, memory subsystems, and diverse data interfaces to support multimodal applications. With a fully integrated on-chip light source, the platform can leverage mature 3D integration technologies, such as through-silicon vias (TSVs), to vertically stack photonic, electronic, logic, and I/O layers, thereby increasing system density and reducing interconnect latency[38]. In parallel, emerging non-volatile memories, such as memristors[39,40] and phase-change materials (PCMs)[41,42] , offer promising routes to reduce static power consumption and minimize storage overhead.

In summary, the QD-MLL-enabled PCU has been demonstrated to improve energy efficiency and system compactness for edge computing. The integrated light source eliminates fiber-based isolators and amplifiers, achieving a 2–6 dB reduction in front-end loss. Co-evaluation with the matched photonic computing link shows >10 dB SNR enhancement and >50% reduction in

thermally induced weight variation. From a system-level perspective, the overall scalability achieves a 1.7× improvement, translating into >40% increase in computational density and ~30% boost in energy efficiency compared to hybrid-source counterparts. Beyond these hardware benefits, the proposed HATF further enables robust and resource-efficient model deployment. Collectively, these advances establish a cross-layer methodology for efficient and reliable photonic computing in next-generation edge AI platforms.

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Author's Contributions

Z.Z. led the design of the ONN links with the assistance of X.O. W.H. fabricated the on-chip QD-MLL under the guidance of Y. W., D.H., H.R., and J.E.B. Z.Z. and W.H. performed the device characterization, assisted by Y.X., Y.S. and X.W. Z.Z and X.W. conducted the device and system modeling. X.W. developed algorithm optimization under the guidance of D.W. H.L. accomplished the design of RF structures under the guidance of A.S. Z.Z. wrote the manuscript with input from others. Y.W. supervised the project. All the authors discussed the data and contributed to the manuscript.

Code and Data Availability

All the data used in this study are available from the corresponding authors upon reasonable request. Source codes of HATF in this study are available online: https://github.com/626619403/MLP_ONN.

Disclosures

The authors declare that they have no competing interests.

Supplementary information

The online version contains supplementary material available at https://doi.org/XXX.X. Correspondence and requests for materials should be addressed to Yating Wan.

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